CHRIST (Deemed to be University)

Department of Computer Science

Master of Computer Applications

Continuous Internal Assessment – III (CIA 3) Examination

Programme	MCA	**Course Title**	Machine Learning
Course Code	MCA 521-4	**Course Type**	Theory and Practical
Credits	4	**Total Marks**	25 Marks

GENERAL INSTRUCTIONS

- This is an end-to-end ML challenge.. Work must demonstrate the complete journey from raw data to an explainable prediction and live demo.
- Submit the complete codebase/notebook, README with execution steps, trained pipeline or reproducible training script, and a three-minute pitch-and-demo video.
- Use an appropriate open/public dataset and cite its source. Do not include personally identifiable or confidential data.
- Any borrowed code, figures, or external assets must be acknowledged. Academic integrity rules apply.

Course Outcome Mapped

CO1: Explain the fundamental principles and scope of Machine Learning

CO2: Implement modelling practices in machine learning for better generalisation

CO3: Interpret predictive modelling results and performance of supervised machine learning techniques

CO4: Assess the outcomes and effects of unsupervised machine learning processes

CO5: Deploy robust machine learning models for interdisciplinary applications

ML FOR SOCIAL GOOD CHALLENGE (25 MARKS)

Mission: Build an end-to-end AI solution that addresses a real human or environmental problem using advanced ensemble machine learning.

The brief - Code with purpose: Machine learning is not only about improving accuracy on abstract datasets; it should create measurable value for people and the planet. Select one mission below and engineer a defensible, reproducible solution.

Mission	Domain	Possible problem

Mission Health	Health	Predict early risk of non-communicable diseases or maternal-health complications in under-resourced clinics.
Mission Earth	Environment	Forecast local AQI, detect forest-fire risk early, or track clean-water accessibility.
Mission Inclusion	Financial inclusion	Predict micro-loan default risk fairly to expand access for underserved populations.
Mission Crisis	Disaster response	Classify and prioritise incoming emergency-help requests during natural disasters.

Q. No.	Questions	CO / RBT	Marks
1	**Real-World Impact Framing** Select one mission domain—Health, Earth, Inclusion, or Crisis. Define the social/environmental problem, intended beneficiaries, prediction target, and measurable impact. Justify why machine learning is suitable, describe the dataset source and unit of analysis, and state responsible-use limitations.	CO5 / L2	**5**
2	**Data Wrangling and Feature Engineering** Build a leakage-safe preprocessing pipeline. Audit and appropriately handle missing values, duplicates, invalid data, outliers, data types, categorical encoding, numerical scaling, and class imbalance. Conduct relevant EDA and engineer defensible domain-informed features. Justify every major preprocessing decision and use a reproducible train/validation/test strategy.	CO2 / L3	**6**
3	**Ensemble Architecture, Tuning, and Comparison** Train a single Decision Tree or Logistic Regression as the baseline. Implement and tune: (a) a bagging model such as Random Forest; (b) a boosting model such as XGBoost, LightGBM, CatBoost, or AdaBoost; and (c) a heterogeneous stacking or voting model. Use cross-validation correctly and prevent leakage in meta-learning. Compare all models on the same untouched test set using suitable metrics such as ROC-AUC, F1, precision, recall, and confusion matrix. Demonstrate whether the best ensemble outperforms the baseline.	CO2, CO3 / L3	**8**

Q. No.	Questions	CO / RBT	Mark s
4	**Model Explainability and Ethics** Use SHAP/LIME or an equivalent defensible method to explain the best model at both global and individual prediction levels. Interpret influential features in domain language and discuss bias/fairness, privacy, uncertainty, false-positive and false-negative costs, human oversight, and limits on deployment.	CO3, CO5 / L5	**4**
5	**Three-Minute Product Pitch and Live Demo** Submit a three-minute video with this sequence: 0:00–0:35—problem and beneficiaries; 0:35–1:15—data cleaning, EDA, and feature engineering; 1:15–2:10—baseline versus bagging, boosting, and stacking/voting results; 2:10–3:00—live prediction on one realistic synthetic record, local explanation, ethics, and limitations. Show the code/repository, a compact results visual, and the live model output.	CO5 / L4	**2**

Required submission: codebase/notebook, README with reproducibility instructions and dataset citation, results/figures, explainability output, ethics statement, and the three-minute pitch-and-demo video.

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	**L2 – Understanding**	**L3 – Applying**
L4 – Analysing	**L5 – Evaluating**	**L6 – Creating**